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Hypoglycemic effects of *Grifola frondosa* (Maitake) polysaccharides F2 and F3 through improvement of insulin resistance in diabetic rats

Chun Xiao,†[a] Qingping Wu,*†[a] Yizhen Xie,[a] Jumei Zhang[a] and Jianbin Tan[b]

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Our laboratory has previously demonstrated that *Grifola frondosa* polysaccharides (GFPs) showed hypoglycemic effects. This study aimed to investigate which polysaccharide-enriched fractions of GFPs were the main active constituents, and to disclose their hypoglycemic mechanism. F2 and F3 were obtained from GFPs and their hypoglycemic effects were investigated. Fasting serum glucose (FSG) levels, fasting serum insulin (FSI) levels and a homeostasis model assessment of insulin resistance (HOMA-IR) were measured, and the hepatic mRNA levels of insulin receptor (IR), insulin receptor substrate-1 (IRS-1), protein tyrosine phosphatase-1B (PTP1B), phosphatidylinositol 3-kinase (PI3K) and Akt/protein kinase B (PKB) were determined by a quantitative polymerase chain reaction (qPCR). The activity of IR and IRS-1 were determined by an enzyme-linked immunosorbent assay (ELISA), and their phospho-protein levels were analyzed with western blotting. F2 and F3 significantly decreased the levels of FSG, FSI and HOMA-IR compared with a diabetic control group ($P < 0.05$). F2 and F3 increased the activity and mRNA levels of IR, and the latter also increased the mRNA levels of IRS-1. As for the protein levels of phospho-IR and IRS-1, both F2 and F3 increased the protein levels of IR (Try 1361), but decreased IRS-1 (Ser307). In the PI3K/Akt pathway, F3 increased the mRNA levels of PI3K and Akt, however, F2 inhibited PTP1B expression. F2 and F3 are presumed to cause an improvement in insulin resistance, triggered by the reactivation of IR and IRS-1.

Introduction

Type 2 diabetes is a global public health crisis that threatens the economies of all nations, particularly those of developing countries.[1,2] In aggregate, there is a pressing need to develop novel modalities for the treatment of diabetes to stem the spread of this global epidemic. Medicinal mushrooms have been valued as a traditional source of natural bioactive compounds over many centuries and have been targeted as potential hypoglycemic and anti-diabetic agents.[3] However, there is insufficient evidence to draw definitive conclusions about the efficacy of individual medicinal mushrooms towards diabetes.[4] Thus, a great deal of interest has been focused on mushrooms for the treatment of diabetes.

Grifola frondosa (maitake) has been confirmed to contain substances with anti-diabetic activity, and it was found to lower blood sugar due to the presence of an α-glucosidase inhibitor.[5] Data suggest that MT-α-glucan has an anti-diabetic effect on KK-Ay mice, which might be related to its effect on insulin receptors (*i.e.*, increasing insulin sensitivity and ameliorating the insulin resistance of peripheral target tissues).[6] A lot of research has focused on the *G. frondosa* SX-fraction, which is a bioactive glycoprotein with a molecular weight of 20 kDa that has exhibited hypoglycemic activity in diabetic mice and in clinical studies of type 2 patients.[7–11] The action of SX is presumed to be associated with the activation of an impaired insulin signal transduction pathway through high glucose levels or under a hyperglycemic milieu, thereby ultimately facilitating glucose uptake.[12] Apart from the fruit bodies, fermented *G. frondosa* rich in vanadium (GFRV) also significantly induced decreases in the blood glucose levels in hyperglycemic mice,[13] and its submerged culture mycelium and broth improved glycemic responses in diabetic rats, with significant decreases in their postprandial blood glucose levels and serum triglyceride levels.[14]

In consistency with the above reports, our laboratory previously demonstrated that *G. frondosa* polysaccharides (GFPs)

[a]*State Key Laboratory of Applied Microbiology Southern China, Guangdong Provincial Key Laboratory of Microbial Culture Collection and Application, Guangdong Open Laboratory of Applied Microbiology, Guangdong Institute of Microbiology, Guangzhou 510070, China. E-mail: Wuqp203@163.com, xiaochun960@hotmail.com; Fax: +86-20-8768-0942; Tel: +86-20-8768-8132*
[b]*Department of Toxicology, Center for Disease Control and Prevention of Guangdong Province, Guangzhou 510020, China*
†Joint first authors.

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showed hypoglycemic effects, lowering the diabetic blood serum glucose level by 50.09% in type 2 diabetic mice induced by streptozotocin (STZ) injection in combination with a high fat diet.[15]

In the present study, our aim was to determine which polysaccharide-enriched fractions in GFPs were the main active constituents and investigate the key hypoglycemic mechanisms. Then, this study investigated the effect of the fractions on the insulin signal pathway to explore possible underlying molecular mechanisms for their action.

Type 2 diabetes is a heterogeneous disorder characterized by hyperglycemia and insulin resistance. Insulin resistance results in the blocking of the insulin signal pathway, and the insulin receptor (IR) and insulin receptor substrate-1 (IRS-1) would be inactivated. The phosphatidylinositol 3-kinase-serine/threonine kinase Akt (PI3K-Akt) pathway plays a pivotal role in insulin signal transduction.[16] The IR is a heterotetrameric glycoprotein consisting of two α-subunits and two β-subunits.[17] The binding of insulin to the α-subunits of IR induces a conformational change that leads to the *trans*-autophosphorylation of tyrosine residues on the β-subunits, activating their tyrosine kinase activity.[18] One such tyrosine residue that is phosphorylated serves as a binding site for the phosphotyrosine binding domains of IR and IRS-1, whose tyrosine residues are then phosphorylated.[19] This tyrosine-phosphorylated IRS-1 acts as a docking site/molecule that binds to and activates PI3K, which in turn activates serine/threonine kinase Akt.[20] Activated Akt ultimately promotes glucose uptake and lipid synthesis.[21] As a negative regulatory factor, protein tyrosine phosphatase-1B (PTP1B) interacts with and dephosphorylates the IR as well as the IRS-1. This is rather a simplified scheme of the insulin signal transduction pathway, which is triggered by activation of the IR (Fig. 1). Thus, one rational approach to overcome such insulin resistance would be by reactivating the IR/IRS-1 to successfully execute the entire signal transduction pathway.

To explore such a mechanism, the effects of F2 and F3 on the mRNA levels, protein activity and phospho-protein levels of IR, IRS-1, PTP1B, PI3K and Akt, which are key elements involved in the insulin signal pathway, were assessed using a quantitative polymerase chain reaction (qPCR), an enzyme-linked immunosorbent assay (ELISA), and western blotting.

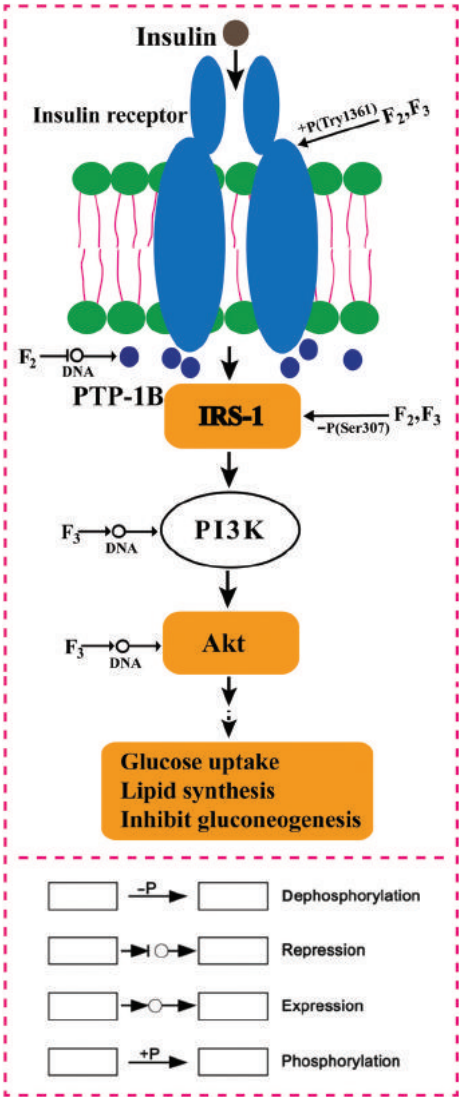

Fig. 1 Action mechanism of F2 and F3 on insulin signal transduction.

Materials and methods

Extraction and purification of polysaccharides

The dried fruiting bodies of *G. frondosa* were homogenized to a fine powder. The powder was mixed with distilled water at a ratio of 1 : 20 (w/v) and extracted at approximately 80 °C. The mixture was filtered and centrifuged at 5000*g* for 10 min at 4 °C. The resulting supernatant was concentrated under a reduced pressure (not exceeding 60 °C) and precipitated with four volumes of absolute ethanol at 4 °C overnight. The resulting precipitate was dispersed in water, dialyzed, and lyophilized to yield the polysaccharides-enriched fraction, named GFPs.

GFPs were applied onto a column of DEAE Sepharose Fast Flow equilibrated with Tris-HCl (10 mmol L^{-1}, pH 8.0), followed by 0.1 M NaCl in Tris-HCl (10 mmol L^{-1}, pH = 8.0), followed by 0.5 M NaCl in Tris-HCl (10 mmol L^{-1}, pH = 8.0) at the same rate. Fractions were assayed for carbohydrate by the phenol–sulfuric acid method[22] and for protein by measuring the absorbance at 280 nm. The concentrations of the fractions were adjusted to 5 mg ml^{-1} or 10 mg ml^{-1}.

Compositional analysis of F2 and F3

The neutral polysaccharide contents of F2 and F3 were determined by the phenol–sulfuric acid method using dextran as the standard.[22,23] The monosaccharide composition was analyzed by gas chromatography-mass spectrometry.[24] The total protein content was determined by the Lowry method using

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bovine serum albumin as the standard.[25] The composition of amino acids in the protein hydrolysate was analyzed by reversed-phase high performance liquid chromatography with a HP1050 analyzer.[26]

Animals

Six-week-old male SD rats (140 ± 20 g) and a standard pellet diet were provided by Guangdong Province Experimental Animals Center (Production Certificate No. scxk (Yue) 2008-0002. Quality Certificate No. 20121209. Experimental Animals License No. syxk (Yue) 2008-0011). The rats had free access to the standard pellet diet and water. The rats were maintained under a constant 12 h light/dark cycle and an environmental temperature of 21–23 °C. All animal procedures complied with the Guide for the Care and Use of Laboratory Animals and were approved by the Ethical Committee of the Center for Disease Control and Prevention of Guangdong Province (Approval ID: 20091224).

Induction of diabetes and assessment of hypoglycemic activity

The rats were adapted to conditions for 3 days and then fed a high-fat diet.[27] After 4 weeks, the rats were fasted overnight before an intraperitoneal (i.p.) injection of freshly prepared STZ [Sigma, 35 mg per kg body weight (BW), dissolved in citrate buffer, pH 4.5]. Rats with fasting serum glucose (FSG) levels >10.0 mmol L^{-1} were considered to be diabetic and were used in the study.

For the experiment, the rats were randomly divided into the following groups (8 rats per group): (1) normal control group, (2) diabetic control group, (3) F2 low dose-treated diabetic group (50 mg kg^{-1} day^{-1}), (4) F2 high dose-treated diabetic group (100 mg kg^{-1} day^{-1}), (5) F3 low dose-treated diabetic group (50 mg kg^{-1} day^{-1}) and (6) F3 high dose-treated diabetic group (100 mg kg^{-1} day^{-1}). The rats of the normal/diabetic control groups were intragastrically administered (i.g.) with saline. All rats were given free access to drinking water and their respective pelleted diet for 14 days. BW gain, food intake and water intake were measured every day. Once a week, the rats were fasted for 5 h and FSG levels were determined.

At the end of the study, the rats were fasted and blood samples were withdrawn from the orbital sinus, after which the rats were anaesthetized and sacrificed by cervical decapitation. The livers were quickly removed, snap-frozen, and stored at −70 °C.

Biochemical analyses

Serum was separated by centrifuging blood samples at 1000*g* for 10 min at 4 °C. FSG levels were determined using a commercially available assay kit (Jiancheng Bioengineering Institute, Nanjing, China) based on the glucose oxidase method.[28] Fasting serum insulin (FSI) levels were determined using a 125I-labled insulin radioimmunoassay kit (Beijing Beifang Biotech Institute, Beijing, China). A homeostasis model assessment-insulin resistance (HOMA-IR) index was calculated to measure the insulin sensitivity of the rats fed experimental diets using the following formula:[29] HOMA-IR = (FSI × FSG) ÷ 22.5. Serum total cholesterol (TC), triglyceride (TG), high density lipoprotein-cholesterol (HDL-c) and low density lipoprotein-cholesterol (LDL-c) levels were measured using commercial kits (BioSino Bio-technology and Science Inc, Beijing, China).

Total RNA extraction

Each liver sample was immediately homogenized in physiological saline with a PRO200 homogenizer (PRO Scientific Inc., Oxford, CT, USA). The total RNA was extracted using an RNAprep Pure Tissue Kit (Tiangen Biotech Ltd, Beijing, China), with a DNaseI digestion step. The total RNA concentration was determined by measurement of the absorbance at 260 nm.

Semiquantitative SYBR real-time PCR

The total RNA was reverse-transcribed using a random-primer first-strand cDNA Kit (Takara, Kyoto, Japan) according to the manufacturer's instructions. PCR was performed in a real-time thermal cycler using an SYBR Green kit (Takara) as follows: 40 cycles of denaturation at 95 °C for 5 s and annealing/elongation at 60 °C for 30 s. The primer sequences of IR, IRS-1, PTP1B, PI3K and Akt are shown in Table 1. The SYBR Green assay was used to detect products from the reverse-transcribed cDNA sample. The mRNA signals were normalized to the GAPDH mRNA signals in each group. The PCR reactions were performed in duplicate for each sample. The relative gene expression was determined as previously described by Livak and Schmittgen.[30]

ELISA analysis of IR and IRS-1 proteins

The activities of IR and IRS-1 in the liver supernatant homogenate were determined using a commercially available ELISA kit (Westang Biotech Co., Ltd, Shanghai, China).

Western blotting analysis of IR (Try1361) and IRS-1 (Ser307)

Liver supernatant homogenate containing 20 mg protein was run on SDS-PAGE and transferred electrophoretically onto the nitrocellulose (NC) membrane. The NC membrane was blocked for 2 h at room temperature and then incubated with anti-IR (Try1361) and IRS-1 (Ser307) polyclonal antibodies (Abcam Inc. Cambridge, MA, USA) overnight at 4 °C, and then with anti-rabbit IgG conjugated with horseradish-peroxidase. Finally, the NC membranes were washed for 30 min with a wash solution, and the immunoreactive lanes on the NC membrane were detected by the enhanced chemiluminescence method and digitalised by the BandScan software version 5.0.

Statistical analysis

All data were expressed as means ± standard deviation (SD). The differences between groups were determined by analysis of variance followed by one-way least significant difference tests. Statistical differences were considered significant at $P < 0.05$. SPSS software version 21.0 was used for all analysis.

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Table 1 Sequences of primers in this study

Gene	Primer sequence		Amplicon (bp)	Accession no.
GAPDH	Forward	5′-CCGCATCTTCTTGTGCAGTG-3′	250	NM_017008.4
	Reverse	5′-TCCCGTTGATGACCAGCTTC-3′		
IR	Forward	5′-TTCATTCAGGAAGACCTTCGA-3′	258	NM_017071.2
	Reverse	5′-AGGCCAGAGATGACAAGTGAC-3′		
IRS-1	Forward	5′-AGAGTGGTGGAGTTGAGTTG-3′	277	NM_012969.1
	Reverse	5′-GGTGTAACAGAAGCAGAAGC-3′		
PI3K	Forward	5′-GAAGGCAACGAGAAGGA-3′	213	XM_008760659.1
	Reverse	5′-CGTCAGCCACATCAAGTA-3′		
Akt	Forward	5′-ACCTCTGAGACCGACACCAG-3′	133	XM_006240631.2
	Reverse	5′-AGGAGAACTGGGGAAAGTGC-3′		
PTP1B	Forward	5′-TGCACAGCATGAGCAGTATG-3′	133	XM_006235639.2
	Reverse	5′-TGTGCCTTTTGTTCCTCC-3′		

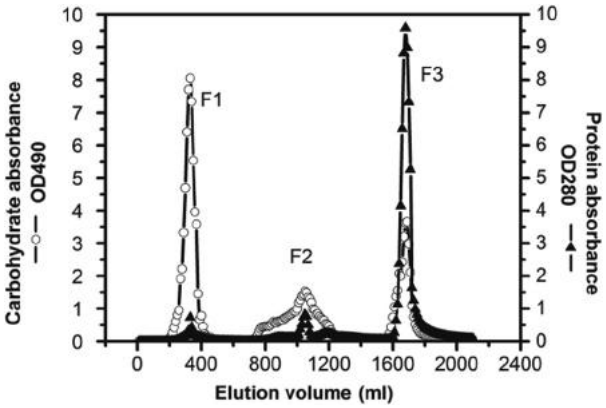

Fig. 2 Fractions of GFPs obtained by DEAE Sepharose Fast Flow chromatography. (Fractions were analyzed by measuring the absorbance at 490 nm for polysaccharides and at 280 nm for the proteins).

Results

Fractionations of the GFPs

The three peak fractions were obtained, as shown in Fig. 2, by DEAE Sepharose Fast Flow chromatography (Fig. 2). Fraction 1 (the first peak) was eluted by Tris-HCl, fraction 2 (the second peak) was eluted by 0.1 mol L^{-1} NaCl in Tris-HCl and fraction 3 (the third peak) was eluted by 0.5 mol L^{-1} NaCl in Tris-HCl. Fraction 2 and fraction 3 were observed to be associated with the protein.

Compositional analysis of F2 and F3

The total polysaccharide and protein content in F2 were 62.5% and 37.5% respectively, and they were 78.3% and 21.7% in F3, respectively (Table 2). It can be seen that glucose, mannose, galactose, xylose, arabinose, rhamnose and ribose were the major monosaccharides of the polysaccharide moieties in F2. However, ribose, arabinose and xylose were the major monosaccharides of the polysaccharide moieties in F3. Sixteen kinds of amino acids constituted the protein moiety. The major amino acids in F2 and F3 were proline, glutamic acid, alanine, arginine, valine, lysine, leucine, histidine, isoleucine, phenylalanine, tyrosine, serine, *etc.*

Table 2 Chemical composition of polysaccharides and proteins in F2 and F3

F2		F3	
Polysaccharides and protein	Composition (%, w w^{-1})	Polysaccharides and protein	Composition (%, w w^{-1})
Polysaccharides (62.5%)[a]		Polysaccharides (78.3%)[e]	
Glucose	26.74[b]	Ribose	74.73[f]
Mannose	22.79	Arabinose	14.20
Galactose	16.76	Xylose	11.08
Xylose	16.02		
Arabinose	14.29		
Rhamnose	2.05		
Ribose	1.35		
Amino acid (37.5%)[c]		Amino acid (21.7%)[g]	
Proline	9.55[d]	Proline	5.84[h]
Glutamic acid	4.26	Alanine	3.59
Alanine	4.14	Glutamic acid	2.01
Arginine	2.90	Arginine	1.44
Valine	2.87	Valine	1.35
Lysine	2.75	Tyrosine	1.22
Leucine	2.53	Leucine	1.20
Histidine	2.16	Phenylalanine	1.10
Isoleucine	1.92	Histidine	1.07
Phenylalanine	1.78	Lysine	1.02
Tyrosine	1.62	Serine	0.97
Serine	1.08	Isoleucine	0.85
Aspartic acid	<0.05	Aspartic acid	<0.05
Glycine	<0.05	Glycine	<0.05
Methionine	<0.05	Methionine	<0.05
Threonine	<0.05	Threonine	<0.05

[a] Percentage of polysaccharides to total F2. [b] Percentage of polysaccharides in F2. [c] Percentage of proteins to total F2. [d] Percentage to total amino acids in F2. [e] Percentage of polysaccharides to total F3. [f] Percentage of polysaccharides in F3. [g] Percentage of proteins to total F3. [h] Percentage to total amino acids in F3.

Effects of F2 and F3 on BW gain, food intake, water intake and food efficiency ratio

From Table 3, the BW gain of diabetic rats administered with F2 (100 mg kg^{-1} day^{-1}) significantly decreased compared with the diabetic control ($P < 0.05$). The daily food intakes of the

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Table 3 Effect of F2 and F3 on BW gain, food intake, water intake and food efficiency ratio in type 2 diabetic rats after treatment for 14 days[a]

Group	Dose (mg kg^{-1} day^{-1})	BW gain (g day^{-1})	Food intake (g g^{-1} day^{-1})	Water intake (ml g^{-1} day^{-1})	Food efficiency ratio[b]
Normal Control	—	4.50 ± 0.60	0.06 ± 0.01	0.10 ± 0.01	0.15 ± 0.02
Diabetic Control	50	3.41 ± 0.48	0.06 ± 0.01	0.21 ± 0.05$^{\alpha\alpha}$	0.17 ± 0.02
F2	50	3.68 ± 0.45	0.07 ± 0.01	0.24 ± 0.07$^{\alpha\alpha}$*	0.15 ± 0.02
F2	100	2.98 ± 0.61*	0.06 ± 0.01	0.21 ± 0.05$^{\alpha\alpha}$	0.14 ± 0.02
F3	50	3.29 ± 0.48	0.07 ± 0.01	0.22 ± 0.06$^{\alpha\alpha}$	0.14 ± 0.02
F3	100	3.41 ± 0.41	0.07 ± 0.02	0.24 ± 0.07*	0.13 ± 0.02

[a] Values are means ± SE (n = 11). [b] BW gain/food intake. *P < 0.05, **P < 0.01, compared to diabetic control rats. $^{\alpha\alpha}P$ < 0.05, compared to control rats.

fraction-administered rats were similar with those of the control rats. The daily water intakes of the diabetic rats were significantly increased compared with those of the normal control rats (P < 0.01), whereas the water intakes of the diabetic rats administered with F2 (50 mg kg^{-1} day^{-1}) and F3 (100 mg kg^{-1} day^{-1}) significantly increased compared with the diabetic control (P < 0.05). The food efficiency ratios representing the BW gain relative to the food intake were the same for the fraction-administered rats, and the ratios of each group were not significantly different.

Effects of F2 and F3 on FSG and FSI levels and HOMA-IR index

As shown in Fig. 3A, there were no significant differences in the FSG levels between the fraction-treated groups and the diabetic control group at the start of the study. After 1 week, however, the FSG level in the F3-treated (50 mg kg^{-1} day^{-1}, 100 mg kg^{-1} day^{-1}) group was significantly lower (P < 0.05) compared with the diabetic control group. The FSG levels in the F2-treated group were also decreased, but the difference was not statistically significant. After 2 weeks, significant decreases of the FSG levels were observed in both of the F2- (100 mg kg^{-1} day^{-1}) and F3-treated (50 mg kg^{-1} day^{-1}, 100 mg kg^{-1} day^{-1}) groups (P < 0.05) compared with the diabetic control group.

At the end of the experiment, the FSI levels were significantly higher in the diabetic control rats than in the normal rats (P < 0.01), consistent with the characteristics of type 2 diabetes (Fig. 3B). However, compared with the diabetic control group, the FSI levels in the fraction-treated groups were significantly lower (P < 0.05). Fig. 3C shows that F2 and F3 significantly lowered the HOMA-IR index (P < 0.01).

Effects of F2 and F3 on fasting serum lipids levels

The plasma lipid levels are usually raised in diabetics, and such an elevation represents a risk factor for coronary heart disease. Hypercholesterolemia and hypertriglyceridemia have been reported to occur in STZ induced diabetic rats and the significant increase observed in our experiment was in accordance with those studies. As shown in Table 4, oral administration of F2 or F3 (50 mg kg^{-1} day^{-1}, 100 mg kg^{-1} day^{-1}) significantly reduced the TC and TG levels in serum compared with the diabetic control rats. We have also observed that oral administration of F3 (50 mg kg^{-1} day^{-1}) significantly decreased the level of LDL-c compared with the diabetic control group. However, no significant change in the HDL-c level was shown in the diabetic control group compared with the normal rats.

Effects of F2 and F3 on the activities of hepatic IR and IRS-1

The activities of IR and IRS-1 were significantly reduced in the diabetic control rats and the lowered levels of IR were enhanced significantly (P < 0.01) in the F2 and F3-treated rats, and the lowered levels of IRS-1 were enhanced significantly (P < 0.01) in the F3-treated rats (Fig. 4).

Effects of F2 and F3 on mRNA levels of IR, IRS-1, PTP1B, PI3K and Akt

The hepatic mRNA levels of IR, IRS-1, PTP1B, PI3K and Akt were similar in the normal and diabetic control groups (Fig. 5). It is interesting to note that F3 increased the mRNA levels of IR, IRS-1, PI3K and Akt compared with diabetic control rats (P < 0.01), and F2 only increased the mRNA levels of IR, IRS-1. As for the negative factor of PTP1B, F2 was capable of decreasing the higher mRNA levels of diabetic rats, but F3 could not do it.

Western blotting analysis of IR (Try1361) and IRS-1 (Ser307)

From Fig. 6, compared with the normal control rats, the IR (Try1361) protein levels in the livers of the diabetic controls were significantly decreased by 90% (P < 0.01). While the diabetic rats were treated with F2 and F3, the IR (Try1361) protein levels in the livers were increased compared with those of the diabetic control rats (P < 0.01). As for IRS-1 (Ser307), compared with the normal control rats, the protein levels in the livers of the diabetic controls were significantly increased by 74% (P < 0.01), but F2 and F3 decreased the protein level (P < 0.01).

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Table 4 Effects of F2 and F3 on TG, TC, HDL-c and LDL-c levels in type 2 diabetic rats

Group	Dose (mg kg^{-1} day^{-1})	TG	TC	HDL-c	LDL-c
Normal	—	1.10 ± 0.35	1.71 ± 0.24	1.17 ± 0.16	0.36 ± 0.07
DM rats	50	6.33 ± 1.88$^{\alpha\alpha}$	6.14 ± 1.92$^{\alpha\alpha}$	1.35 ± 0.27	1.91 ± 0.68$^{\alpha\alpha}$
F2	50	3.62 ± 0.46*	3.95 ± 1.51	1.32 ± 0.22	1.71 ± 0.28
F2	100	3.60 ± 0.82*	3.24 ± 1.27*	1.23 ± 0.16	1.33 ± 0.26
F3	50	2.52 ± 1.09**	2.52 ± 1.01**	1.13 ± 0.15*	0.80 ± 0.46*
F3	100	4.01 ± 0.95*	3.74 ± 0.69*	1.20 ± 0.27	1.59 ± 0.42

Values represent means ± SD (n = 8). *P < 0.05, **P < 0.01 *vs.* diabetic control and $^{\alpha\alpha}P$ < 0.01 *vs.* normal group. TG: triglyceride, TC: total cholesterol, HDL-c: high density lipoprotein-cholesterol, LDL-c: low density lipoprotein-cholesterol.

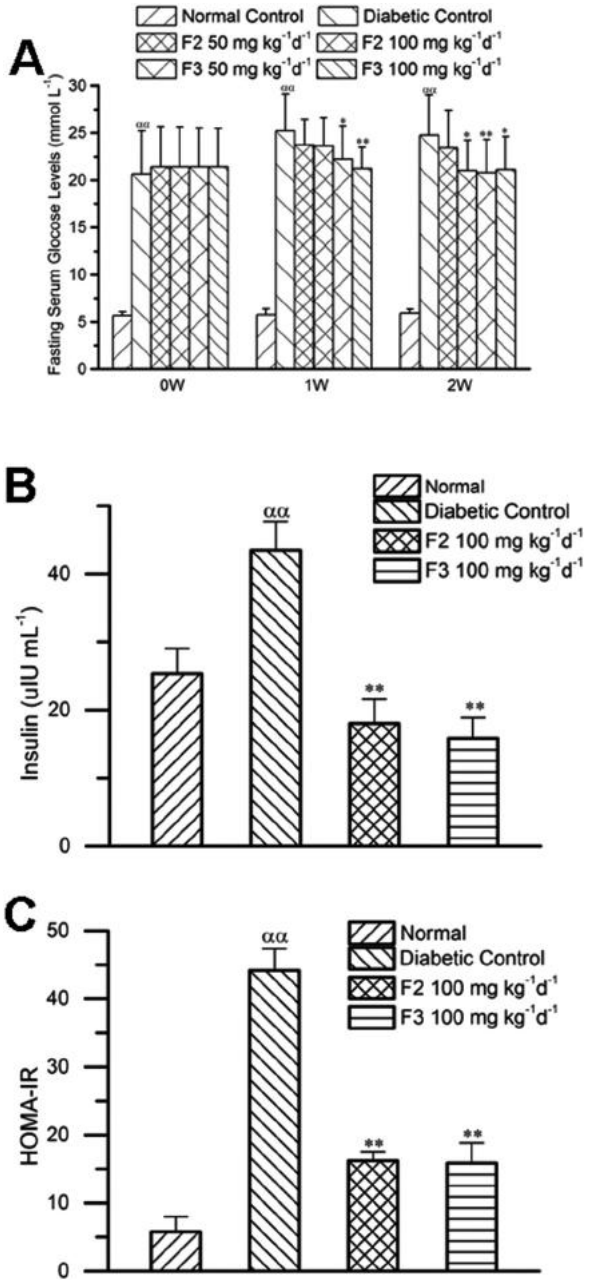

Fig. 3 (A) Effects of F2 and F3 on FSG levels, (B) effects of F2 and F3 on fasting insulin levels, (C) effects of F2 and F3 on the HOMA-IR index in diabetic rats. (Values represent the means ± SD (n = 8). *P < 0.05, **P < 0.01 *vs.* diabetic control and $^{\alpha\alpha}P$ < 0.01 *vs.* normal group.)

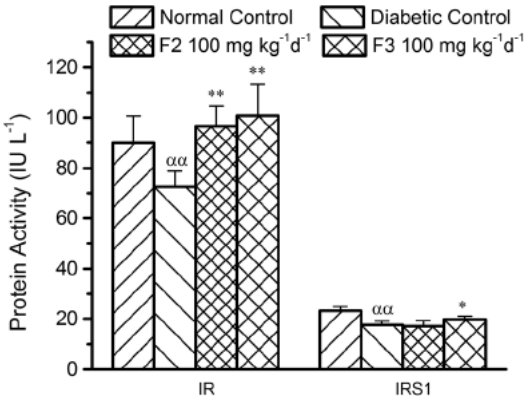

Fig. 4 Effects of F2 and F3 on the protein activity of hepatic IR and IRS-1. (Values represent means ± SD (n = 8). *P < 0.05, **P < 0.01 *vs.* diabetic control and $^{\alpha\alpha}P$ < 0.01 *vs.* normal group.)

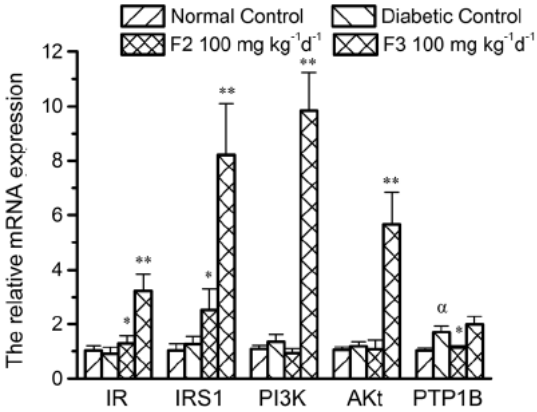

Fig. 5 Effects of F2 and F3 on the mRNA expression of IR, IRS-1, PI3K, Akt and PTP1B. (Values represent means ± SD (n = 8 per group). *P < 0.05, **P < 0.01 *vs.* diabetic control and $^{\alpha\alpha}P$ < 0.01 *vs.* normal control.)

Discussion

The hypoglycemic agents from mushroom were almost all polysaccharides or polysaccharide-protein/peptide complexes.[31,32] Lei found that *G. frondosa* MT-α-glucan has an anti-diabetic effect on KK-Ay mice.[33] *G. frondosa* SX-fraction is a

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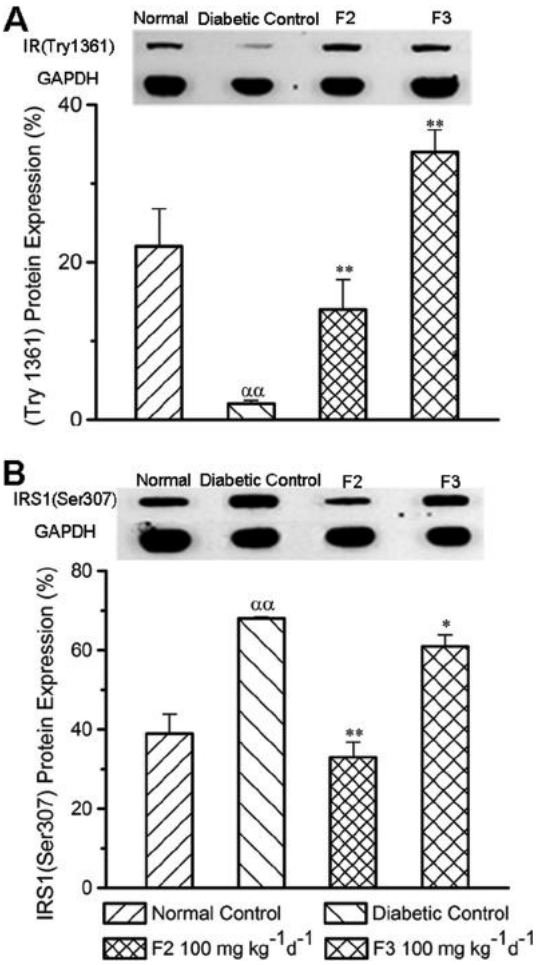

Fig. 6 (A) Effects of F2 and F3 on protein expression of IR (Try1361), (B) effects of F2 and F3 on protein expression of IRS-1 (Ser307). (Values represent means ± SD (n = 8 per group). *P < 0.05, **P < 0.01 *vs.* diabetic control and $^{\alpha\alpha}P$ < 0.01 *vs.* normal control.)

bioactive glycoprotein with a molecular weight of 20 kDa that has exhibited hypoglycemic activity in diabetic mice and in clinical studies of type 2 patients.[7–9,11,34] As for *Ganoderma lucidum*, ganoderan A was composed of rhamnose, galactose and glucose with a molecular weight of 23 kDa, ganoderan B was composed of protein, mannose, glucose and galacturonic acid with a molecular weight of 7.4 kDa, and ganoderan C was composed of glucose and galactose with a molecular weight of 5.8 kDa.[35–37] In consistency with the previous report, our lab successfully isolated *G. lucidum* polysaccharide F31, which seemed to be a polysaccharide-protein/peptide complex with a weight-average molecular weight of 15.9 kDa. In the present study, we isolated two fractions of polysaccharide-protein/peptide complexes from *G. frondosa.* The structures of the main chain and branches need to be studied further. To date, the relationship between the structural features and the anti-hyperglycemic activity is not clear. So, a further analysis study on the F2 and F3 polysaccharide structures would contribute towards the understanding of the structure–activity relationship.

In this study, diabetes in rats was induced by the injection of STZ in combination with high-fat dietary feeding, and the character of hyperglycemia, hyperinsulin and insulin resistance was similar to type 2 diabetes. In a state of insulin resistance, glucose uptake and utilization are dramatically decreased, and skeletal muscle becomes metabolically inflexible and unable to switch between glucose and fatty acid use.[38] We found that F2 and F3 lowered FSG and FSI levels and improved insulin resistance in diabetic rats (Fig. 3). The results were consistent with our previous report that GFPs lowered hyperglycemia by counteracting insulin resistance.[15] As we know, type 2 diabetes is related to significant cardiovascular morbidity and mortality by modulation of lipid profiles. Dyslipidemia, which occurs in approximately 50% of patients with type 2 diabetes, results in cardiovascular complications through elevated triglyceride levels, low levels of HDL-c, and a high rise in LDL-c levels.[39] Lipids play an important physiological role in skeletal muscle, the heart, liver and pancreas. Deregulation of fatty acid metabolism is the main culprit in the development of insulin resistance and type 2 diabetes.[40] In the present study, associated with the corrected glycemia, TC, TG and LDL-c levels in the plasma were reduced during the period of F2 and F3 administration (Table 4). So the F2 and F3-lowered FSG levels, associated with the lowered lipid accumulation in the serum, may be a result of an improvement in insulin sensitivity.

As an important indication that F2 and F3 improve insulin resistance, this study investigated the effect of the fractions on the insulin signal pathway to explore the possible underlying molecular mechanisms. The PI3K-Akt pathway plays a pivotal role in insulin signal transduction.[16] The liver is a major site of glucose metabolism in response to insulin.[41] So the key components of IR, IRS-1, PTP1B, PI3K and Akt in the liver, which play a pivotal role in the insulin signaling pathway, were investigated in the present study.

Insulin acts by binding to its cell surface receptor, thus activating the IR's intrinsic tyrosine kinase activity, resulting in IR autophosphorylation and the phosphorylation of several substrates. Tyrosine phosphorylated residues on the receptor itself and on subsequently bound receptor substrates provide docking sites for downstream signalling molecules.[42] While the phosphorylation of IRS-1 on the tyrosine residue is required for insulin-stimulated responses, the phosphorylation of IRS-1 on serine residues has a dual role, either to enhance or to terminate the insulin effects.[21] However, the serine phosphorylation of IRS-1 at Ser307 is a negative regulatory process.[43] That is to say, the serine phosphorylation of IRS-1 (Ser307) would impair its tyrosine phosphorylation, then attenuate the rest of the PI3K-Akt signal pathway, whereas the tyrosine-phosphorylated IR would facilitate the pathway. In this study, we found that the IR (Try1361) phosphorylation levels decreased and the IRS-1 (Ser307) phosphorylation levels increased in diabetic rats (Fig. 6). While the diabetic rats were treated with F2 and F3, the IR (Try1361) protein levels in the

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livers were increased and the IRS-1 (Ser307) phosphorylation levels decreased. That is to say, such inactivation of IR and IRS-1 was reversed or reactivated by F2 and F3, presumably aiding the occurrence of successive signaling events. These results suggest that the signal pathway being impaired with a high level of blood serum glucose could be reactivated or turned on by F2 and F3.

Actually, activated IRS is responsible for the activation of PI3K, which in turn promotes the serine/threonine phosphorylation of Akt for its activation. Akt is also required for the insulin regulation of gluconeogenesis and glucose release in the liver. Interestingly, we found that the mRNA levels of PI3K and Akt in the insulin signaling transduction were significantly improved in the F3-treated rats (Fig. 5). It is presumed that the PI3K-Akt pathway was activated by F3.

Our findings that F2 and F3 improved resistance are in accordance with *G. frondosa* SX. The hypoglycemic action of SX is presumed to be associated with the activation of such an impaired insulin signal transduction pathway involving IR, IRS, and Akt, thereby ultimately facilitating glucose uptake.[12]

Protein tyrosine phosphatase 1B (PTP1B) plays an important role in the negative regulation of the insulin signal transduction pathway and has emerged as a novel therapeutic strategy for the treatment of type 2 diabetes.[44,45] PTP1B interacts with and dephosphorylates the IR as well as the IR substrate (IRS). If PTP1B were over expressed, then most of the IRS would be dephosphorylated and a series of enzymes, such as PI3K and Akt, participating in the process of glucose uptake would be inactivated since the insulin transduction pathway is blocked. Recent gene knockout studies in mice identify PTP1B as a promising target for anti-diabetes/obesity drug discovery.[46] That is to say, the inhibition of PTP1B has emerged as an attractive therapeutic strategy to treat type 2 diabetes and obesity. In this study, it is interesting that F2 inhibited the mRNA levels of PTP1B significantly ($P < 0.05$) but F3 did not decrease the levels (Fig. 5). There is little reported about a PTP1B inhibitor from edible fungi. A novel PTP1B activity inhibitor named Fudan-Yueyang-Ganoderma luciden (FYGL),[47–50] screened from *G. lucidum*, also demonstrates anti-diabetic properties in animal models of type 2 diabetes.

The present study showed that F2 and F3 increased the activity and the mRNA levels of IR and IRS-1 (Fig. 5). F2 and F3 increased the protein expression of IR (Try1361) but decreased the protein expression of IRS-1 (ser307) (Fig. 6). In the PI3K/Akt pathway, F3 increased the mRNA levels of PI3K and Akt, however, F2 inhibited the mRNA levels of PTP1B. These results suggest that the *Grifola frondosa* polysaccharide fractions F2 and F3 may specifically target the insulin signal pathway, and, in particular, F3 may activate the IR and IRS therein that trigger the subsequent PI3K/Akt signaling events; however, F3 inhibited the mRNA levels of PTP1B. Their improvement of the insulin resistance action mode is summarized in Fig. 1.

However, as for functional foods, safety is the most important factor. From Table 3, the daily food intakes and food efficiency ratios of the fraction-administered rats were similar with those of the control rats. The intragastrical (i.g.) administration of F2 and F3 would not result in any adverse effects and poor food intake; furthermore, the hypoglycemic effect of F2 and F3 was not due to poor food intake. These data suggested that F2 and F3 may be non-toxic to diabetic rats.

Conclusions

In conclusion, F2 and F3 decreased FSG levels through the improvement of insulin sensitivity as a result of the increased protein levels of phospho-IR (Try 1361) and decreased levels of phospho-IRS-1 (Ser307). These results demonstrate that *G. frondosa* polysaccharides are useful as potential functional food ingredients for the prevention and treatment of type 2 diabetes.

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